Free Body Diagram

- ❑ IDEALIZATIONS
- ❑ 2D SUPPORT REACTIONS
- ❑ FREE BODY DIAGRAM

Idealizations

PARTICLE

Idealizations are used in mechanics in order to simplify application of the theory

A particle has mass, but its size can be neglected

Idealizations

RIGID BODY

Idealizations are used in mechanics in order to simplify application of the theory

A rigid body is a combination of a large number of particles in which all the particles remain at a fixed distance from one another, both before and after applying a load

Idealizations

CONCENTRATED FORCE

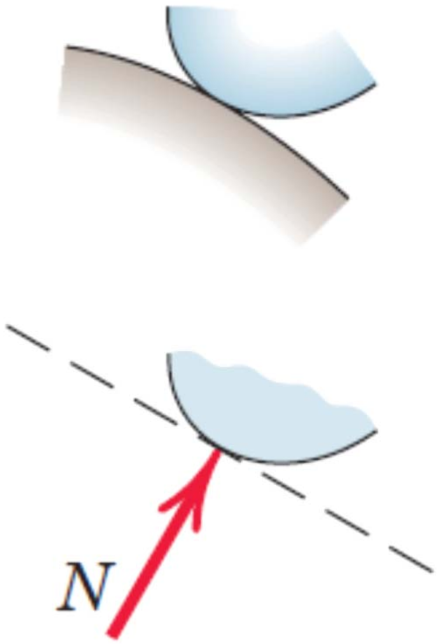
Idealizations are used in mechanics in order to simplify application of the theory

A Concentrated Force is assumed to act at a point on a body

2D Support Reactions

SMOOTH SURFACES

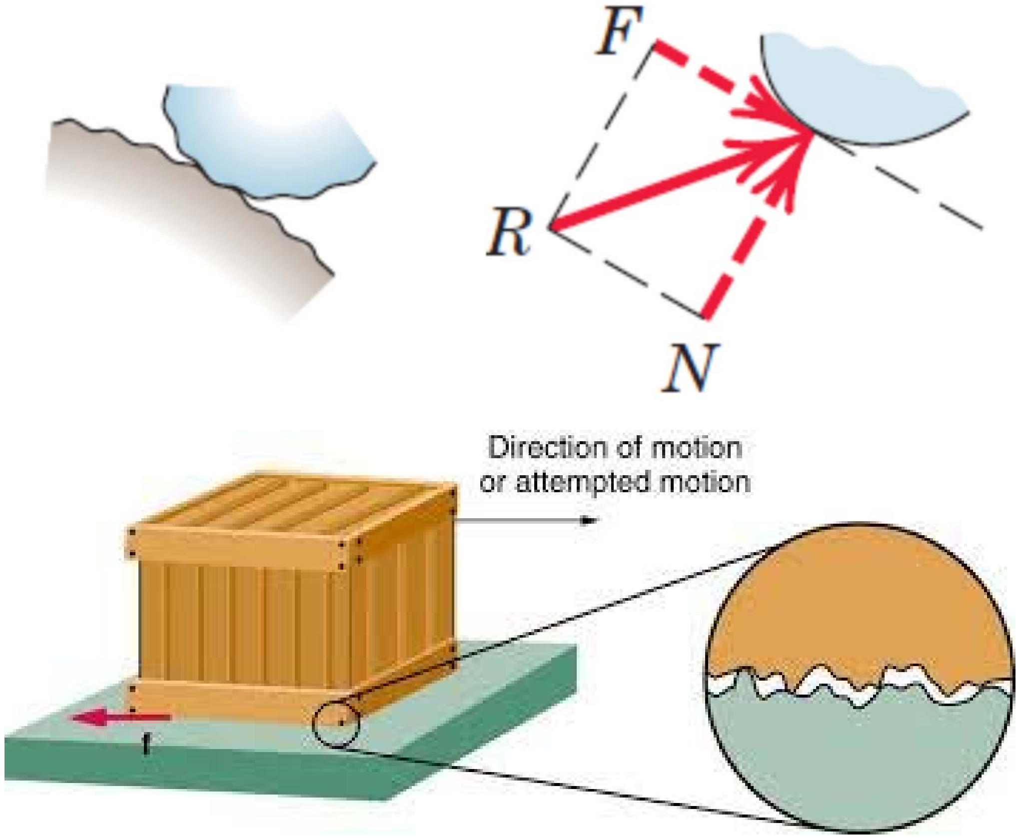
Contact force is compressive and is normal to the surface



2D Support Reactions

ROUGH SURFACES

Rough surfaces are capable of supporting a tangential component F (frictional force) as well as a normal component N of the resultant contact force R.



2D Support Reactions

ROLLER SUPPORT

Roller, rocker, or ball support transmits a compressive force normal to the supporting surface.

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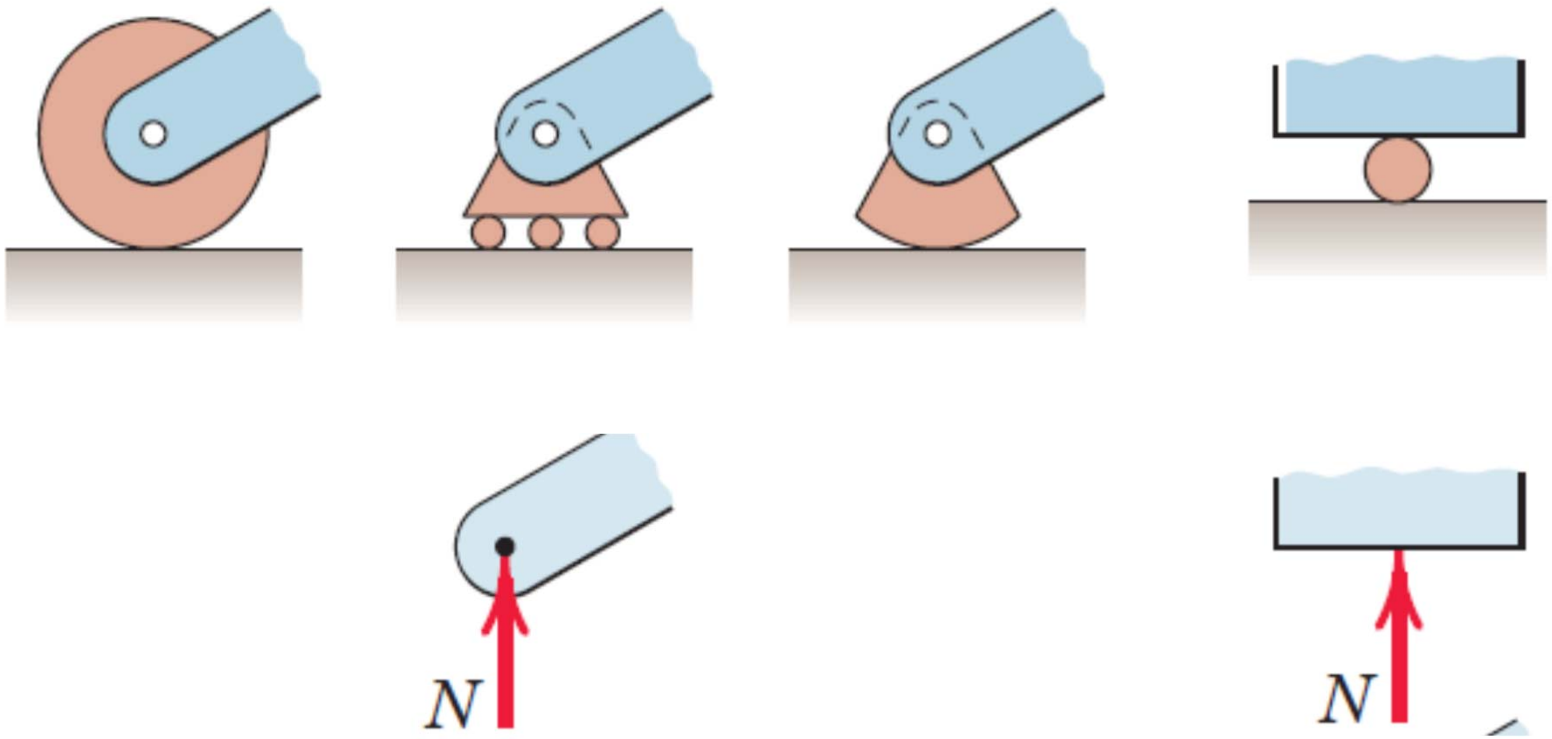

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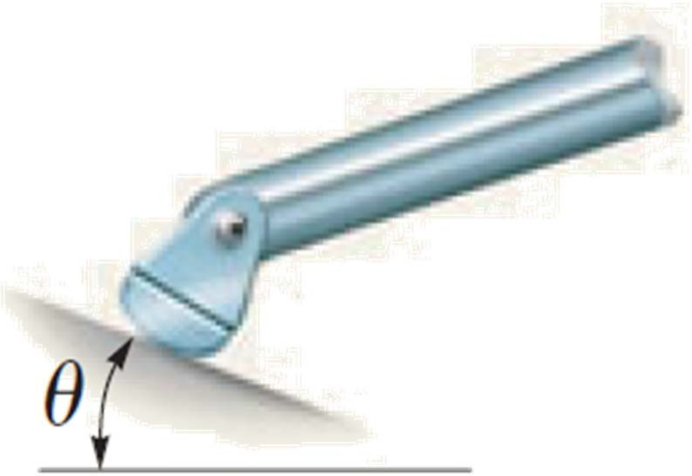
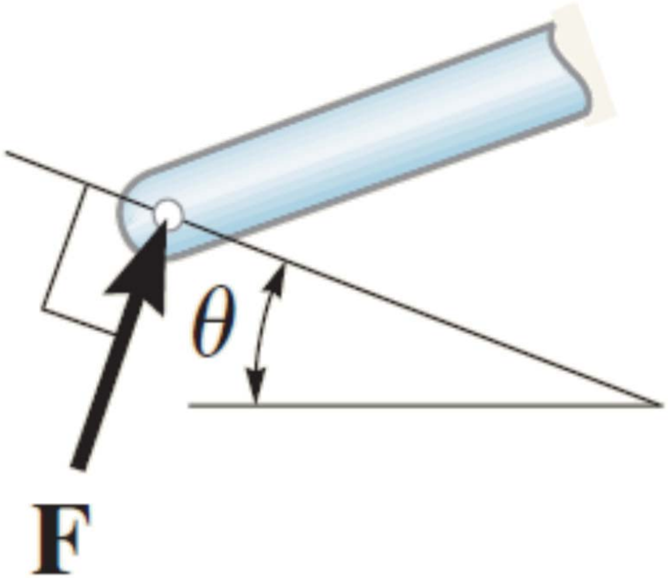
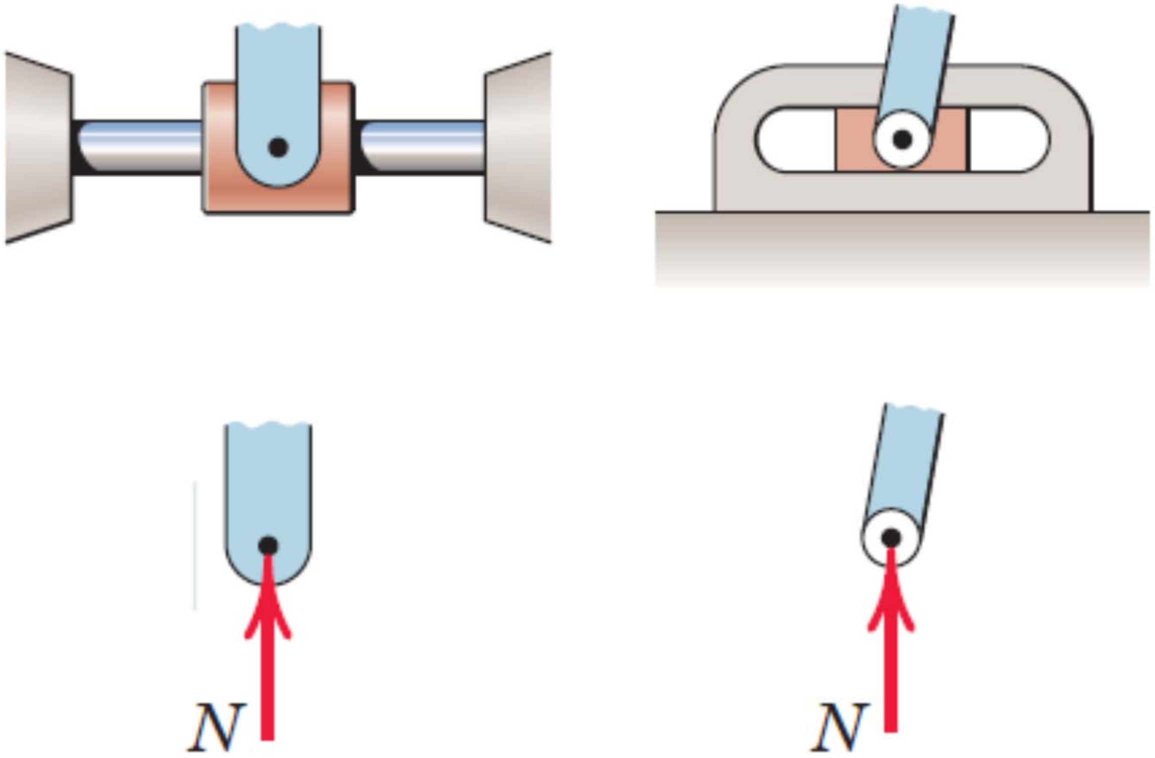




2D Support Reactions

FREELY SLIDING GUIDE

Collar or slider free to move along smooth guides; can support force normal to guide only.



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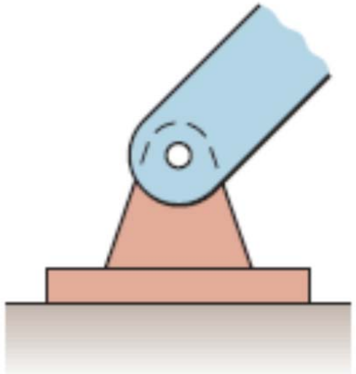
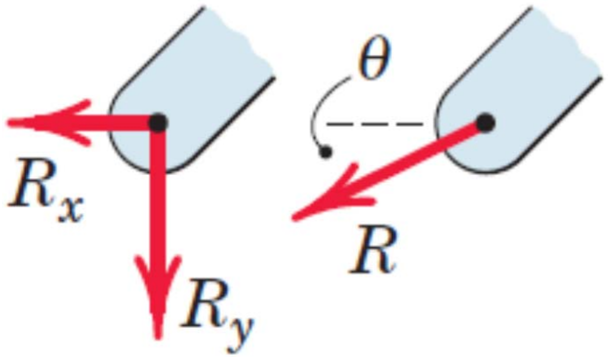
2D Support Reactions

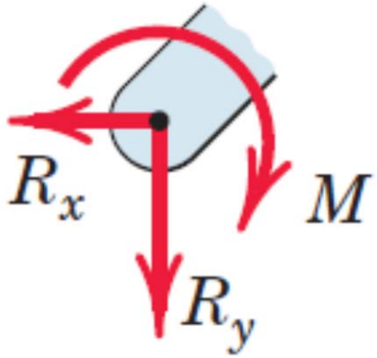
PIN CONNECTION

A freely hinged pin connection is capable of supporting a force in any direction in the plane normal to the pin axis.





A pin not free to turn also supports a couple M.



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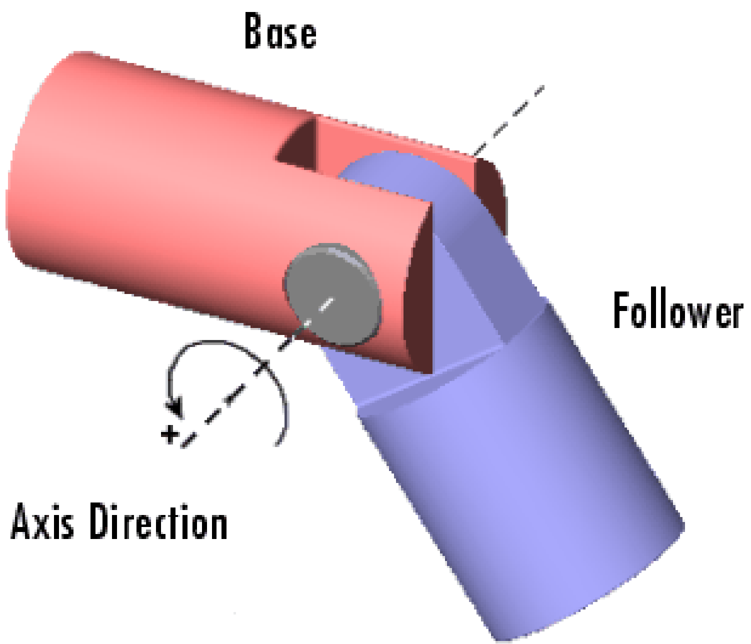

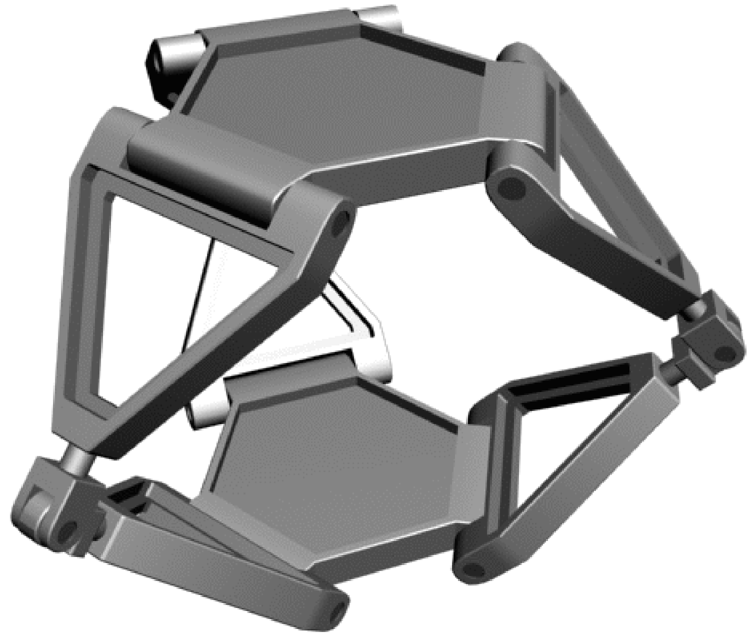

2D Support Reactions

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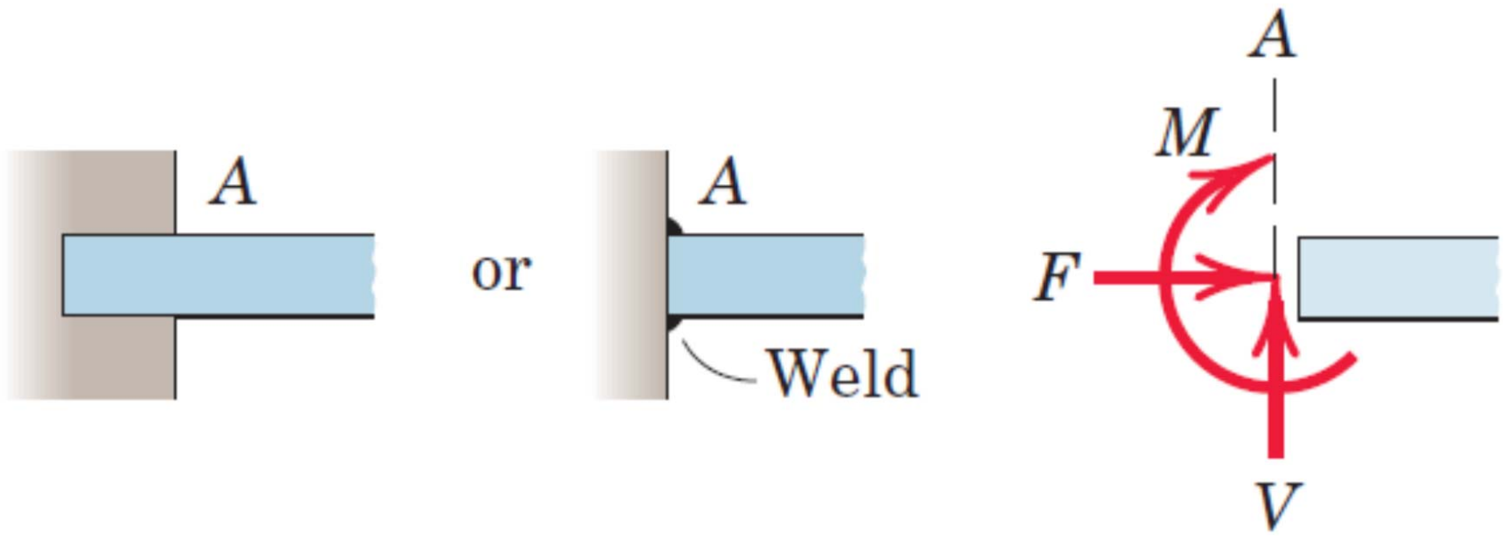
2D Support Reactions

BUILT-IN OR FIXED SUPPORT

A built-in or fixed support is capable of supporting an axial force F, a transverse force V (shear force), and a couple M (bending moment) to prevent rotation



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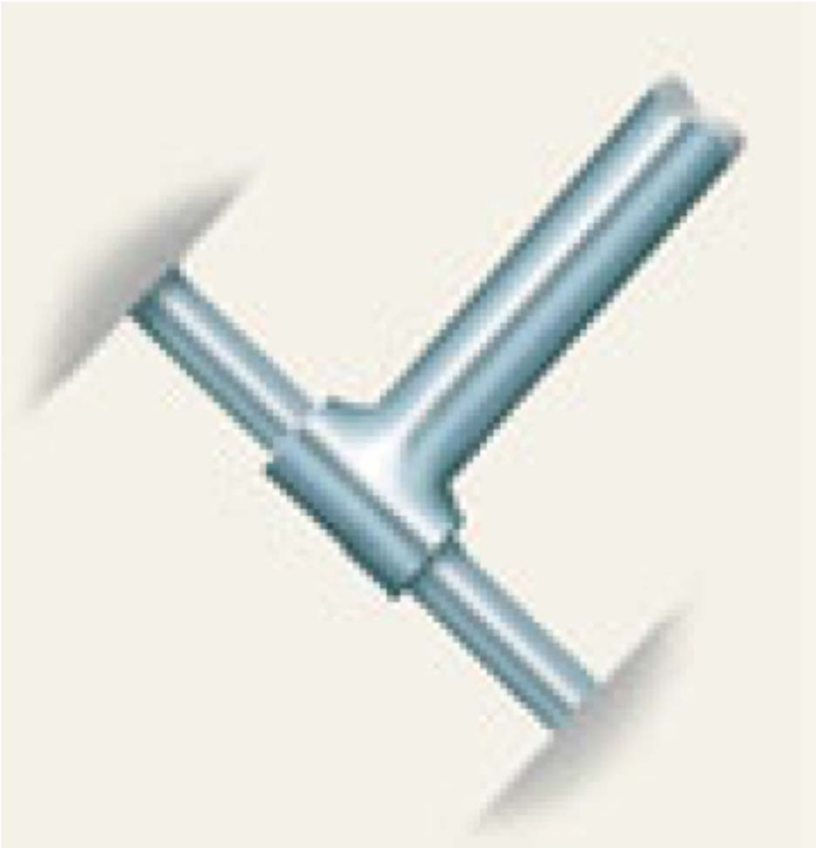
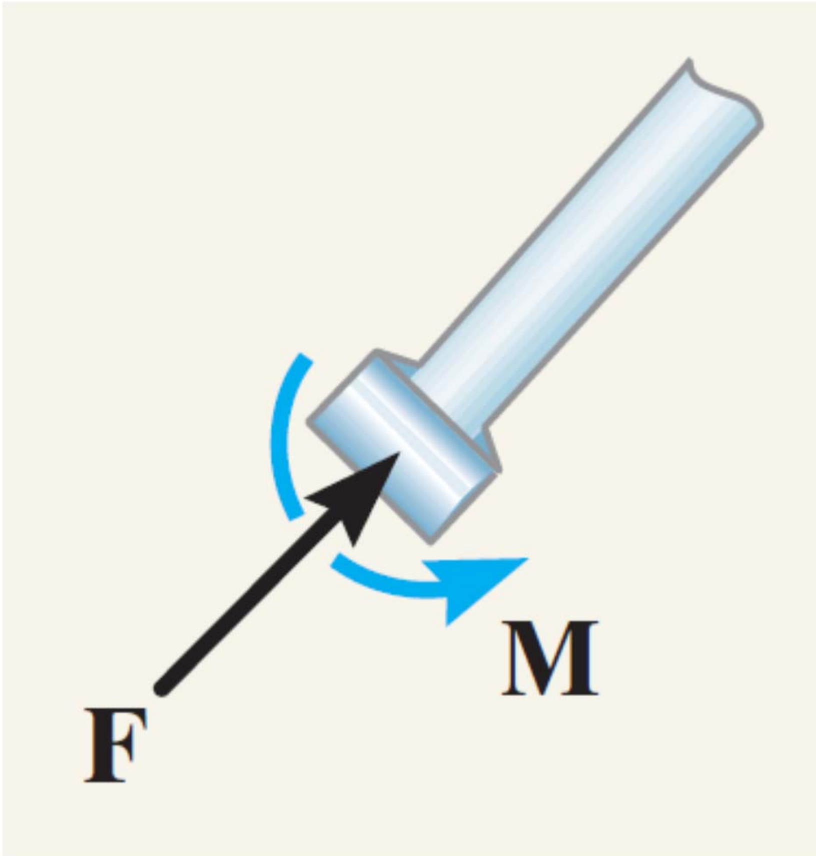
2D Support Reactions

MEMBER FIXED CONNECTED TO COLLAR

The reactions are the couple moment and the force which acts perpendicular to the rod.

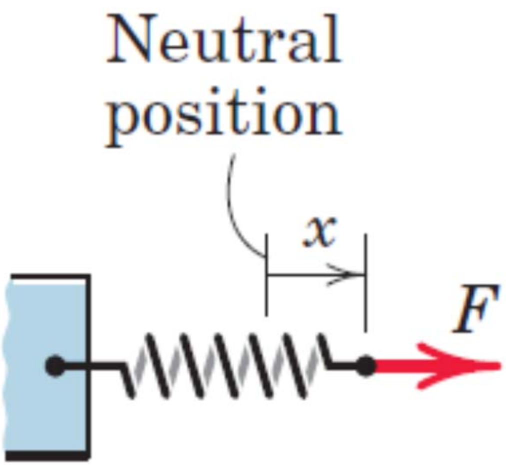


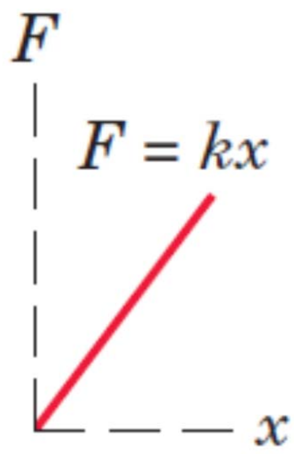


2D Support Reactions

SPRING ACTION

For a linearly elastic spring the stiffness k is the force required to deform the spring a unit distance.

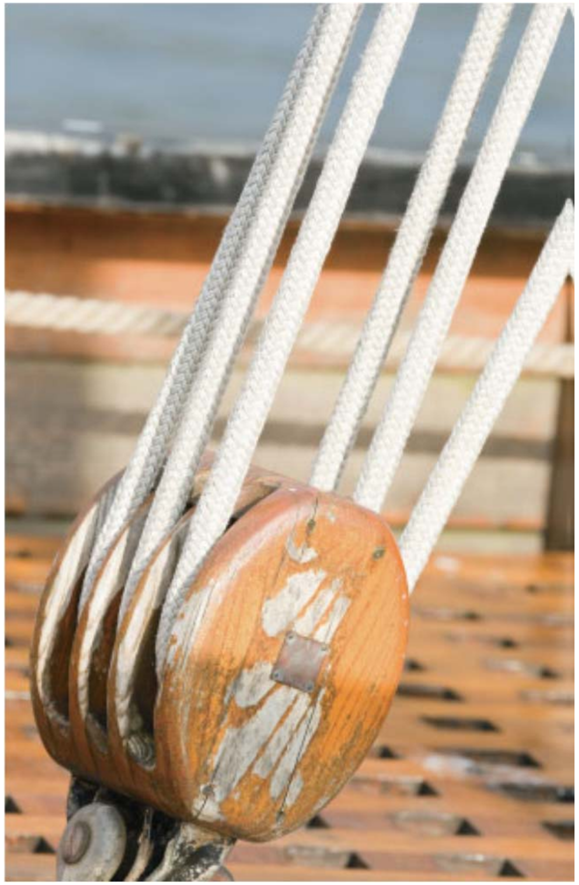




2D Support Reactions

Flexible cable, belt, chain, or rope

Force exerted by a flexible cable is always a tension away from the body in the direction of the cable.



2D Support Reactions

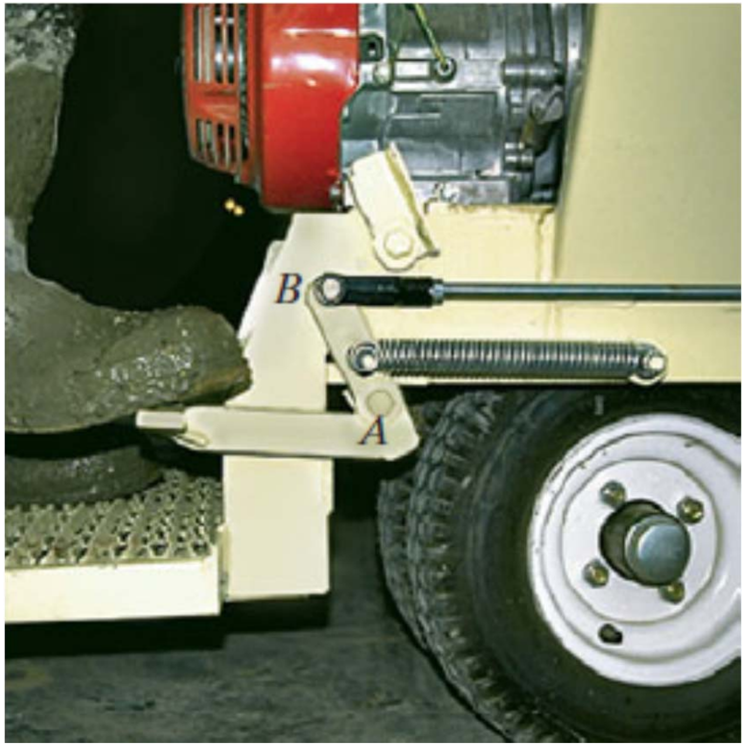
GRAVITATIONAL ATTRACTION

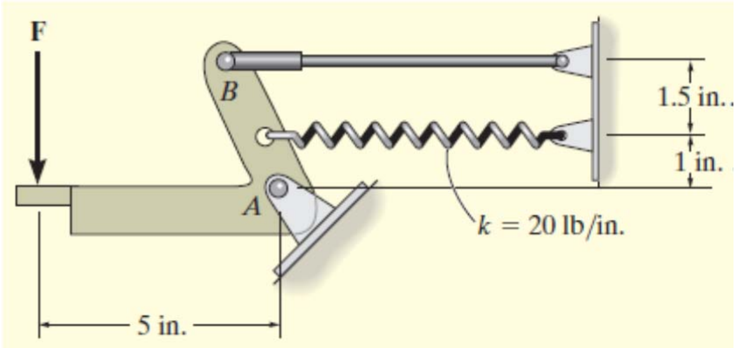
The resultant of gravitational attraction on all elements of a body of mass m is the weight W = mg and acts toward the center of the earth through the center mass G.

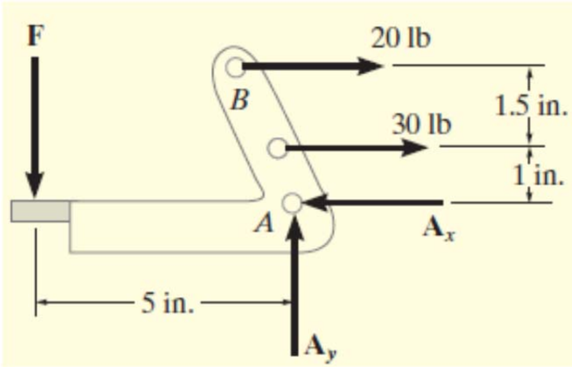
Free Body Diagram

INTRODUCTION

Specification of all the known and unknown <u>external</u> forces that act on the body







A thorough understanding of how to draw a free-body diagram is of primary importance for solving problems in mechanics

Free Body Diagram

PROCEDURE

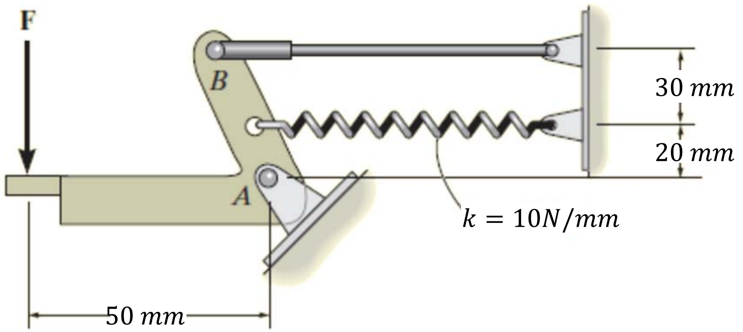
1 – Establish Coordinate Axis

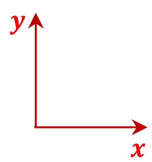
2 – Isolate Body & Draw Outlined Shape

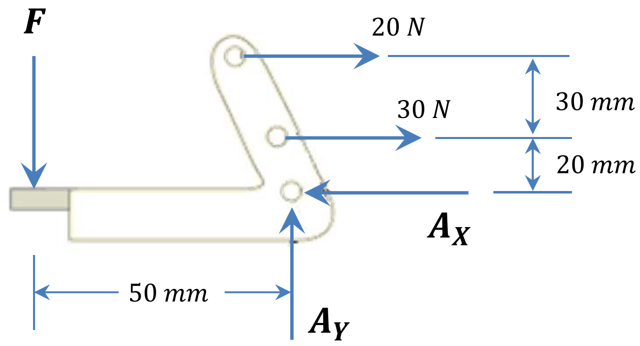
3 – Draw All Applied Forces and Moments

4 – Identify Support Reactions

5 – Indicate Dimensions Necessary for Computing

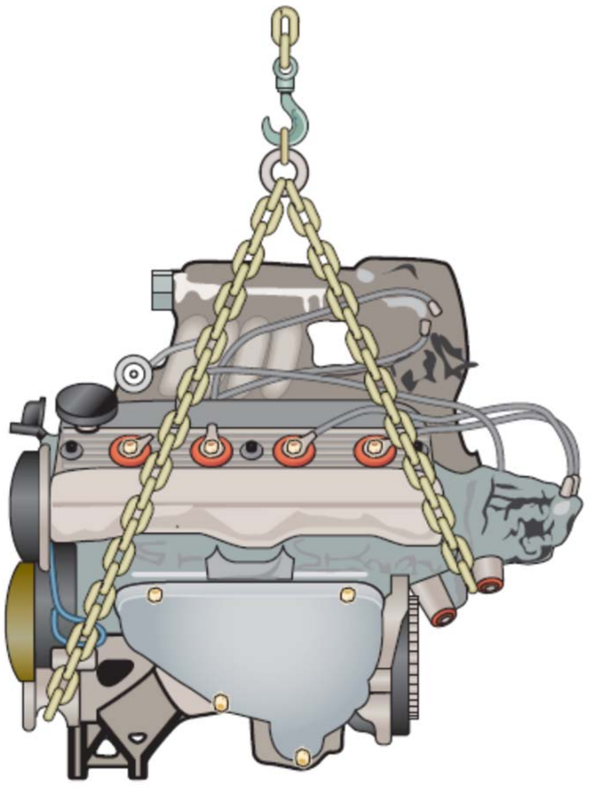


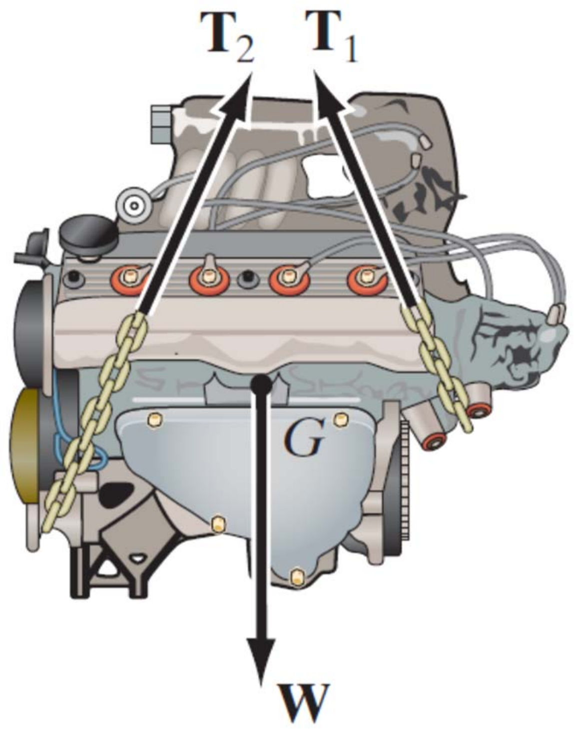




Free Body Diagram

EXAMPLE I

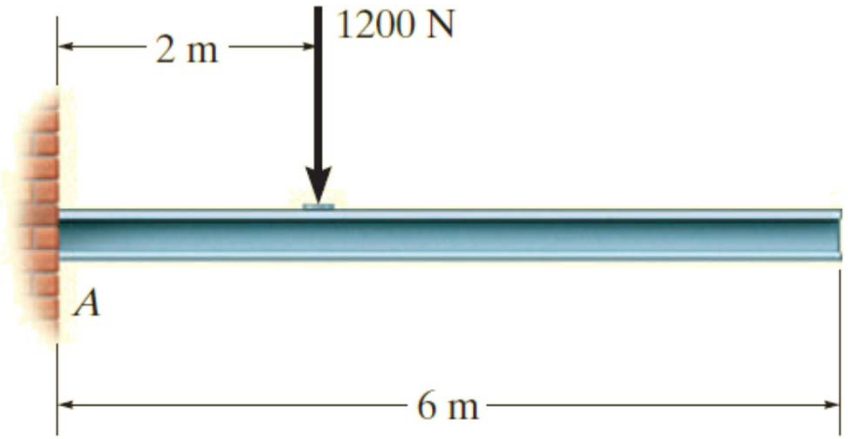




Free Body Diagram

EXAMPLE II

Draw the free-body diagram of the uniform beam. The beam has a mass of 100 kg.



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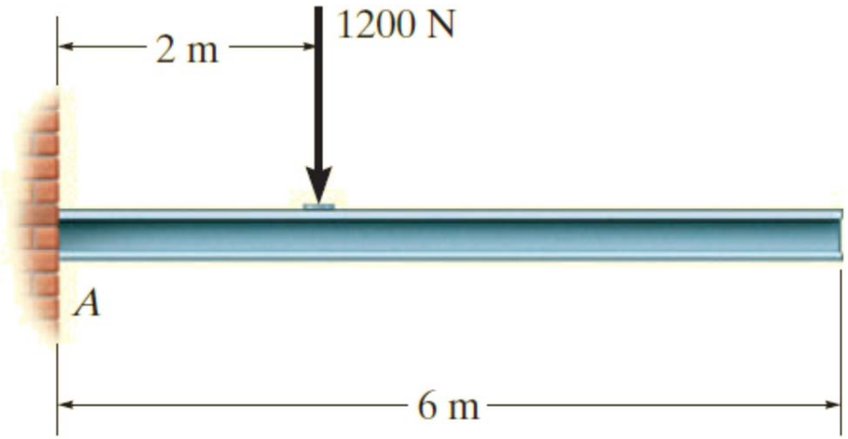
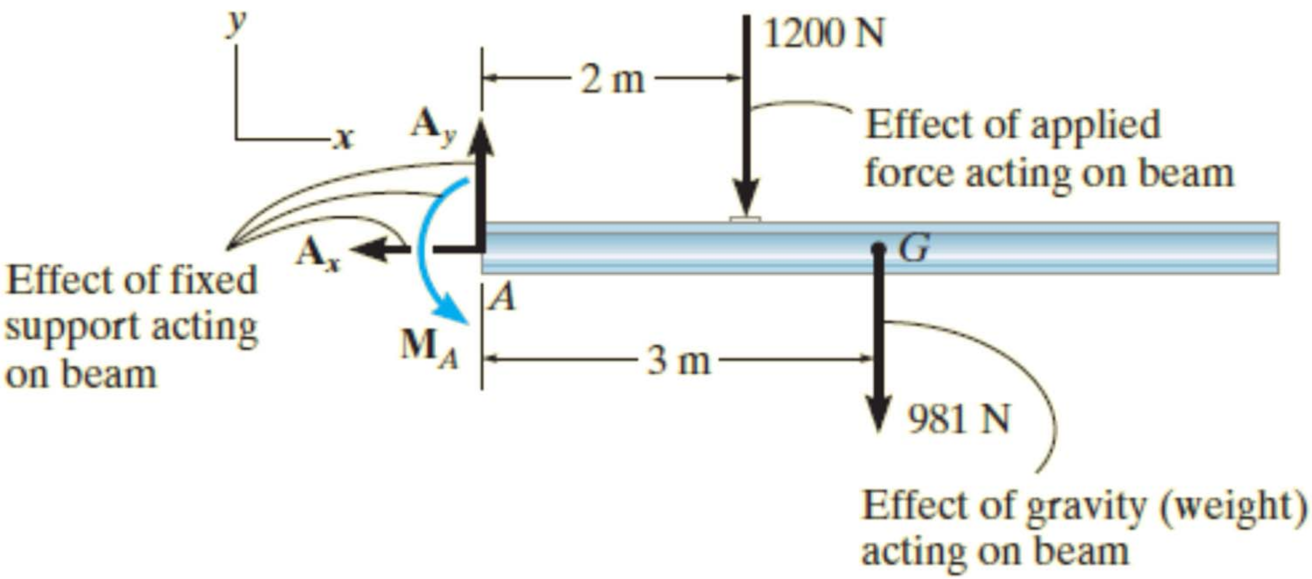
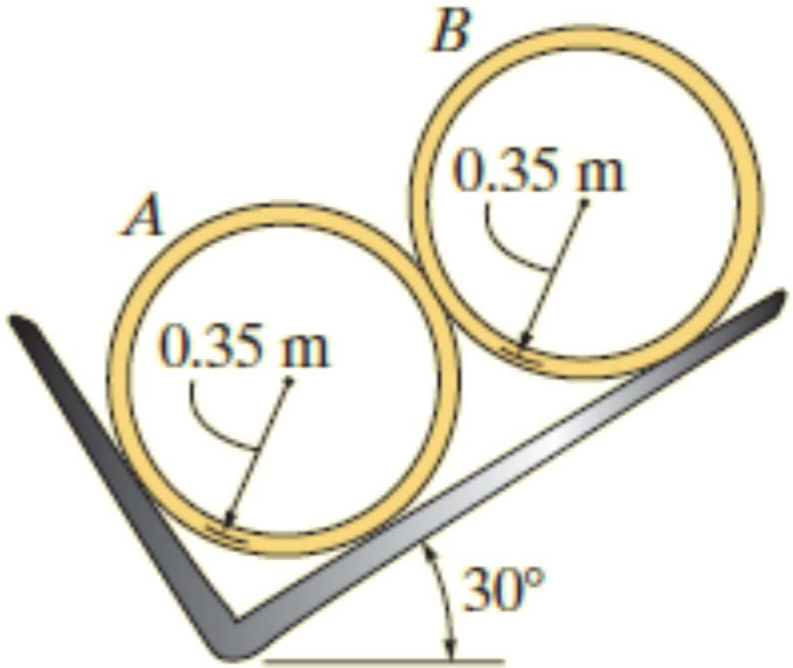




Free Body Diagram

EXAMPLE III

Two smooth pipes, each having a mass of 300 kg, are supported by the forked tines of the tractor in Fig. 5–9a. Draw the free-body diagrams for each pipe and both pipes together.



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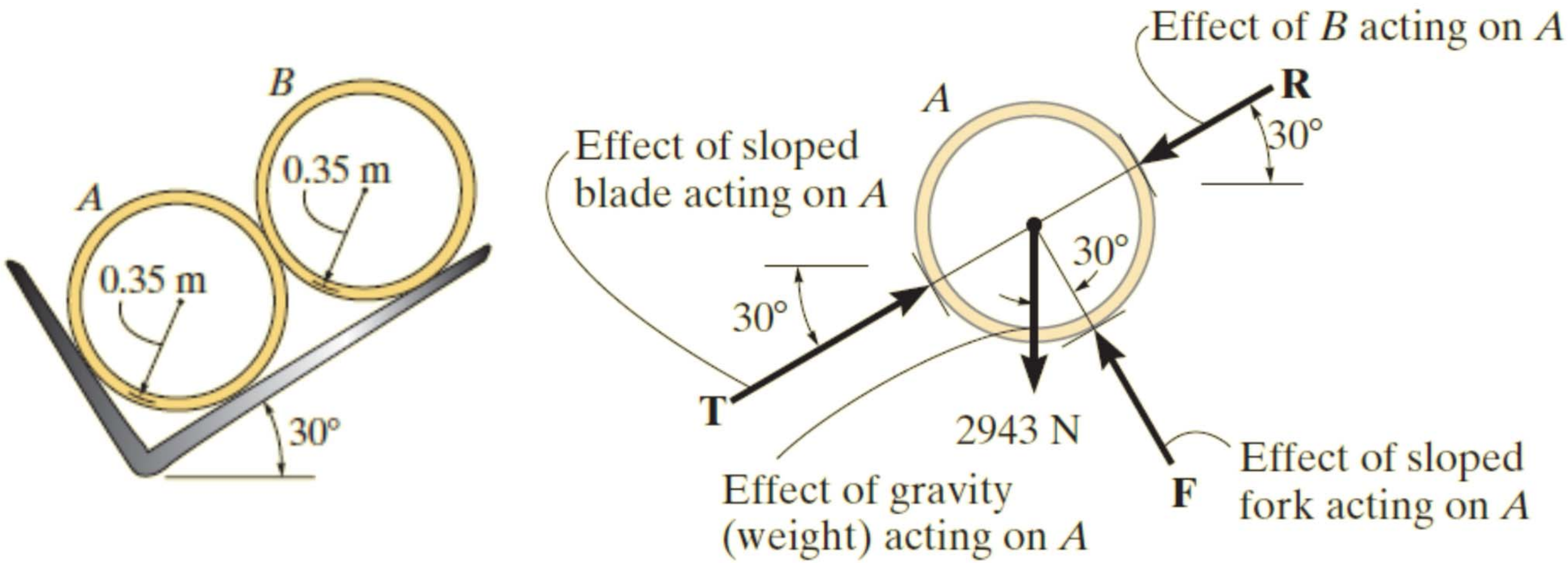

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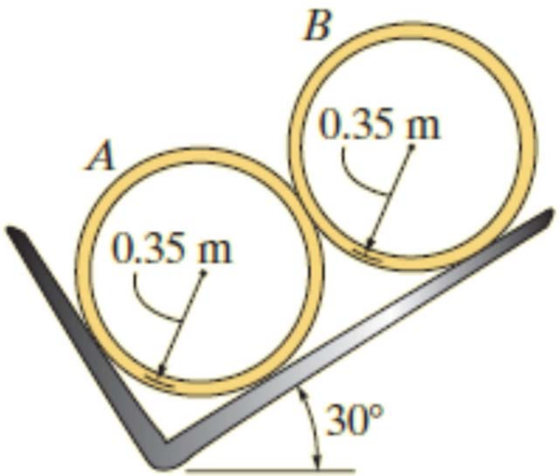
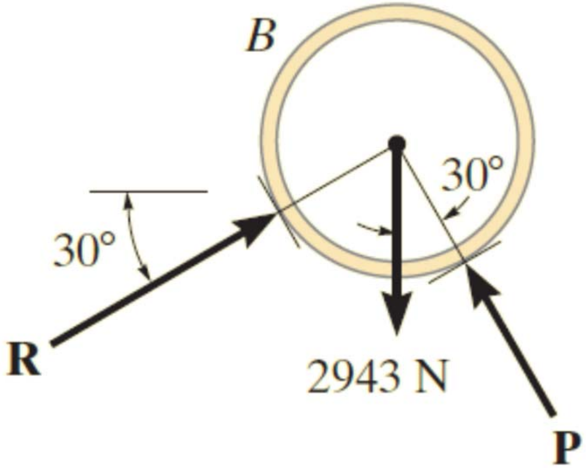
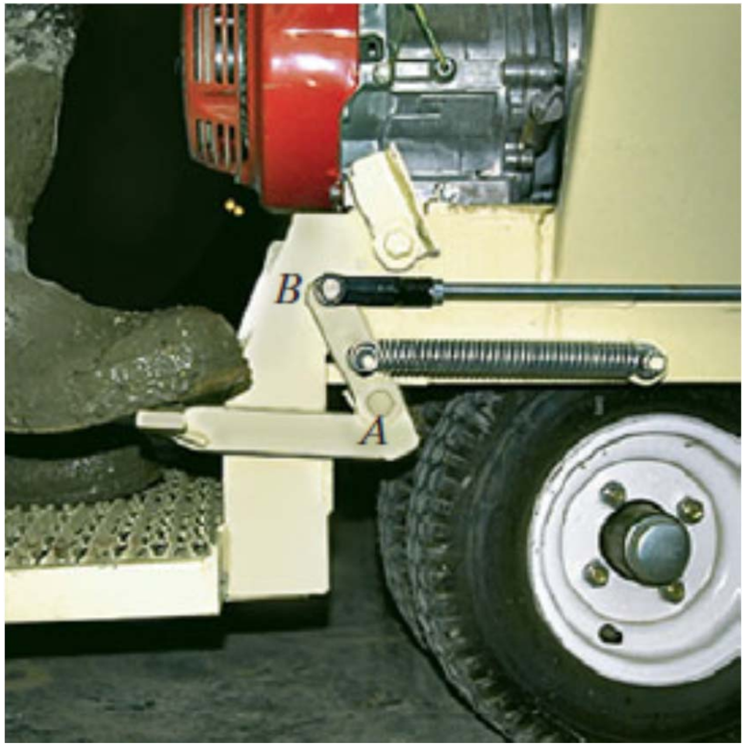


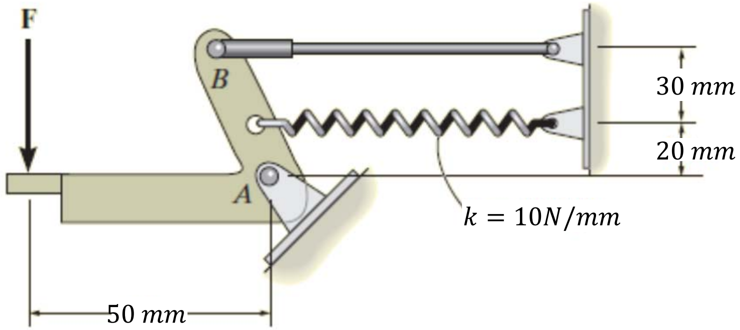


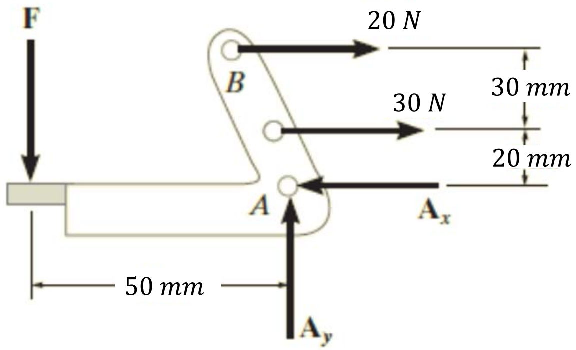
Free Body Diagram

EXAMPLE IV

Draw the free-body diagram of the foot lever. The operator applies a vertical force to the pedal so that the spring is stretched 3 mm. and the force in the short link at B is 20 N.



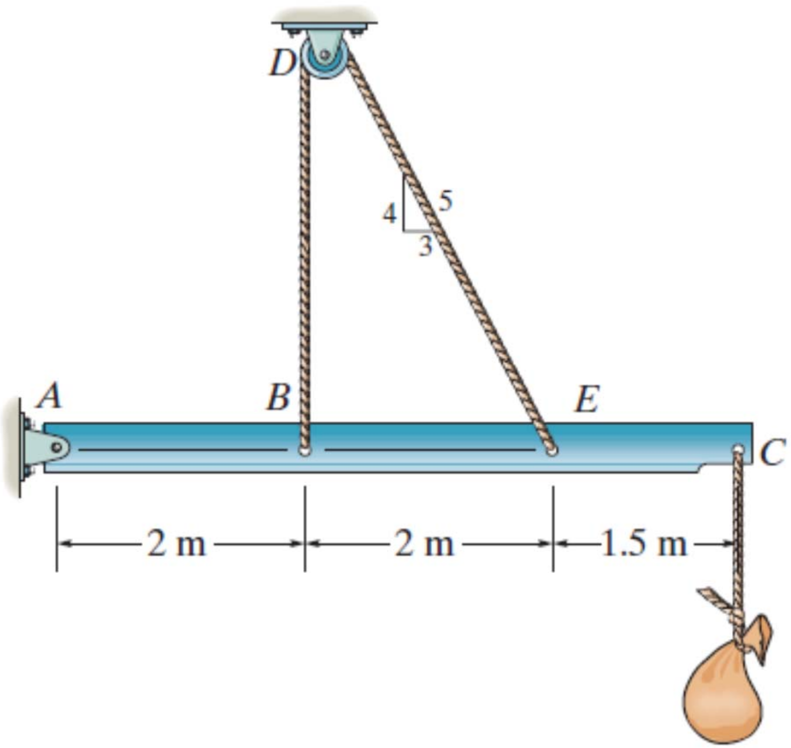




Free Body Diagram

EXAMPLE V

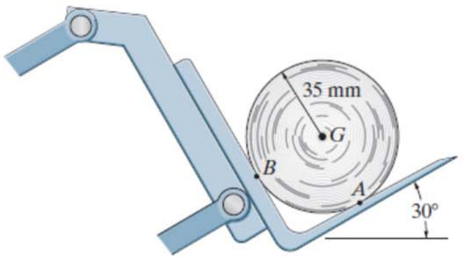
Draw the free-body diagram of the beam which supports the 80-kg load and is supported by the pin at A and a cable which wraps around the pulley at D.



Free Body Diagram

ASSIGNMENT # 1

Problems 5.1 to 5.6



Prob. 5–1